

Assignment #1

Instructor: Guillaume St-Onge

Due 7 October 2026, 11:59 p.m.

Teams of 3–4 students

Worth 8% (CSI 4124) • 5% (SYS 5110)

Total: 60 points

Numerical integration and characterization of compartmental models

Compartmental models partition a population into a small number of classes and describe how individuals flow between them. They are very flexible and commonly used to model epidemic spreading and social dynamics, such as information diffusion and behaviour adoption.

While the number of individuals in a given class is a *discrete* quantity, in the limit of a large population, the dynamics for the *average* number of individuals in different classes is well-defined and can be studied using differential equations. Here we study classic compartmental models in this “large population regime”, hence all references to a “number” of individuals below should be thought of as averages.

Part 1 — The SIS model

In the Susceptible–Infectious–Susceptible (SIS) model, an individual is either susceptible or infectious, and recovery confers no immunity. It is arguably the simplest compartmental model available. Let $s(t)$ and $i(t)$ be the *numbers* of susceptible and infectious individuals in a population of fixed size $N = s + i$. Their evolution is described by the ODEs

$$\frac{ds}{dt} = -\beta \frac{si}{N} + \alpha i, \quad \frac{di}{dt} = \beta \frac{si}{N} - \alpha i, \quad (1)$$

where $\alpha > 0$ is the recovery rate and $\beta > 0$ the transmission rate. The interpretation of the term si/N is that susceptible and infectious individuals interact with one another at rate $1/N$ (e.g., random encounters) and the disease is transmitted at rate β .

The ratio $\mathcal{R}_0 = \beta/\alpha$ is called the *basic reproduction number*, and corresponds to the average number of secondary infections caused by an infectious individual before they recover. For the sake of simplicity, we take $\alpha = 1$ throughout, which amounts to measuring time in units of the infectious period.

Because $s + i = N$ is conserved, the pair reduces to a single equation,

$$\frac{di}{dt} = (\mathcal{R}_0 - 1)i - \mathcal{R}_0 \frac{i^2}{N}. \quad (2)$$

For $\mathcal{R}_0 \neq 1$ and $i(0) = i_0 > 0$, Eq. (2) has the closed-form solution

$$i(t) = \frac{K i_0 e^{(\mathcal{R}_0 - 1)t}}{K + i_0 (e^{(\mathcal{R}_0 - 1)t} - 1)}, \quad K = N \left(1 - \frac{1}{\mathcal{R}_0} \right), \quad (3)$$

which you will use as the reference for measuring numerical error.

Unless a question says otherwise, use

$$N = 10^5, \quad i_0 = 10, \quad \mathcal{R}_0 = 1.5. \quad (4)$$

A1.1 Integrating the SIS model (25 points)

- a. Implement the explicit Euler, RK2 and RK4 integrators as seen in class. They should be functions that take a vector field (right-hand side of the ODE system), an initial condition, a horizon T and a number of steps M , and return the trajectory $\{i_j\}_{j=0}^M$ together with the times $\{t_j\}_{j=0}^M$. The step size is then fixed implicitly by $h = T/M$, which guarantees that the last point of the trajectory lands exactly on T . Integrate Eq. (2) to $T = 40$ with $M = 4000$ (that is, $h = 10^{-2}$), using all three integrators. Produce two plots: the three trajectories against Eq. (3), and the absolute error $|i_j - i(t_j)|$ against t on a semi-logarithmic scale. Comment on what the second plot shows that the first one does not. (10 points)

Hint. For each integrator, write a separate function for a *single* step of size h .

- b. **Local error.** The local error is the error committed by a *single* step of an integrator. Starting from a point $i(t_0)$ taken from Eq. (3) (the exact solution), take one step of size h with the three integrators and compare the result with $i(t_0 + h)$. Do this for $t_0 \in \{2, 10, 20\}$, with step sizes $h = 20/M$ where M runs over roughly log-spaced integers from 100 to 2000, so that h runs from 0.2 down to 10^{-2} . For each t_0 , plot the error against h on log-log axes (one curve per integrator) and obtain the slope with `scipy.stats.linregress` for the relationship

$$\log \epsilon = a \log(h) + b. \quad (5)$$

Interpret the slopes you obtain: what do they tell you about the three integrators, and does the result depend on t_0 ? (10 points)

Hint. `np.round(np.logspace(np.log10(100), np.log10(2000), 10)).astype(int)` gives such a set of integers M .

- c. **Global error.** Now integrate to a fixed horizon $T = 20$ and measure the error at that time only,

$$\epsilon(h) = |i_M - i(T)|. \quad (6)$$

Use the same values of M as in part b. Plot $\epsilon(h)$ against h on log-log axes, extract the slope the same way, and compare the three slopes with those of part b. Does this match your expectation? (5 points)

You will use **RK4** with an appropriate value of M for the rest of the assignment.

A1.2 From prevalence to observed cases (15 points)

Equation (2) tracks $i(t)$, the number of people infectious *right now*. That is not what a public health agency reports. Suppose a fraction $p = 0.10$ of all new infections are identified and recorded, and let $C(t)$ be the cumulative number of cases reported up to time t .

- a. Write $C(t)$ as an integral from 0 to t , and state the extra differential equation for C that you can append to Eq. (2) so that a single call to your integrator returns both i and C . (5 points)

Hint. The integrand is the rate at which susceptible individuals *become* infectious.

- b. Integrate the augmented system with RK4 to $T = 40$ and plot $i(t)$ and $C(t)$. The curve $C(t)$ has two clearly distinct regimes: identify them and explain both qualitatively by comparing with the behaviour of $i(t)$ over the same interval. (10 points)

Part 2 — The SIRS model

Immunity is rarely permanent. Let us consider an additional class: recovered individuals. Let $r(t)$ be the number of recovered individuals. Recovered individuals become susceptible again at rate $w > 0$, which gives us the Susceptible–Infectious–Recovered–Susceptible (SIRS) model

$$\frac{ds}{dt} = -\mathcal{R}_0 \frac{si}{N} + wr, \quad \frac{di}{dt} = \mathcal{R}_0 \frac{si}{N} - i, \quad \frac{dr}{dt} = i - wr, \quad (7)$$

with $s + i + r = N$. Because the population size is preserved, the third variable is redundant: substituting $r = N - s - i$ leaves a two-dimensional system,

$$\frac{ds}{dt} = -\mathcal{R}_0 \frac{si}{N} + w(N - s - i), \quad \frac{di}{dt} = \mathcal{R}_0 \frac{si}{N} - i. \quad (8)$$

A1.3 Phase portrait and behaviour of the SIRS model (20 points)

- a. Integrate Eq. (8) for $\mathcal{R}_0 = 1.5$ and $w = 0.1$ from several different initial conditions. Plot $i(t)$ against t , and in a separate panel, draw the phase portrait—your trajectories in the (s, i) plane, with the vector field in the background. Describe qualitatively what you observe. (10 points)
- b. Push the waning rate towards higher values, $w \in \{1, 10, 100\}$, and overlay $i(t)$ with the SIS trajectory of Part 1 for the same N, i_0 and $\mathcal{R}_0$. What do you observe, and what does it tell you about the relevance of the recovered compartment when immunity wanes very fast? (10 points)

What to hand in

Hand in one Jupyter notebook (*.ipynb) per team in the Brightspace assignment box, together with a PDF export of the executed notebook and an AI Use Declaration (see below).

The notebook is the report. It must contain everything a reader needs to judge your results: the problem set-up, the Python code you ran, the figures, and your conclusions. Presentation matters — use sections, label your axes, and cite any reference outside of the course material.

Put the names of all team members at the top of the notebook, and name the file A1_ followed by the team members' last names, separated by underscores (e.g. A1_name1_name2_name3.ipynb).

Use of artificial intelligence

AI assistance is permitted for this assignment, subject to the documentation requirements below. In contemporary software development, artificial intelligence is increasingly used by programmers to improve productivity, a trend that is likely to continue. Learning to use these tools critically and responsibly is therefore a valuable professional skill. Nevertheless, you must thoroughly understand the concepts and methods on which you are evaluated, particularly because AI tools are not available during in-person evaluations. You remain responsible for verifying, understanding, and being able to explain everything in your submission.

Permitted, properly declared AI use does not reduce your mark. Every submission must contain an **AI Use Declaration**, even if you did not use AI assistance.

1. **Did you use a generative AI tool while completing this assignment?** Answer Yes or No. This includes any type of assistance with code, writing, or editing.

If you answer **No**, include the following confirmation and omit the remaining questions:

I confirm that I did not use generative AI in preparing this submission.

If you answer **Yes**, answer the remaining questions:

2. **How did you use generative AI?** Select all that apply:
 - clarifying concepts or checking your understanding;
 - brainstorming or planning an approach;
 - writing or revising code;
 - debugging code or interpreting error messages;
 - interpreting data, results, tables, or plots;
 - writing or revising explanations;
 - other uses, which you must specify.
3. **Briefly identify the sections or tasks for which AI assistance was used and describe how it influenced your work.**
4. **How did you verify the AI-generated suggestions, code, or explanations?**
5. **Include the following confirmation:**

I confirm that this declaration is complete and that I understand and can explain everything in my submission.